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GG-S-931

NOVEMBER 6, 1947

FEDERAL SPECIFICATION

SYRINGES; GLASS, WITH BULB

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than February 1, 1948, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A–1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification:

ZZ–R–601—Rubber-Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).

A–2. For specifications and publications applicable only to individual departments, see section H.

B. TYPES, STYLES, AND CAPACITIES

B–1. Glass syringes with bulb, in accordance with this specification, shall be of the following types, styles, and capacities, as specified in the invitation for bids:

Type I.—Glass tip.
- Style A.—Cone tip.
  - Capacities:
    - ⅛ ounce.
    - ¼ ounce.
    - ½ ounce.
    - 1 ounce.
- Style B.—Tapered tip.
  - Capacities:
    - ½ ounce.
    - 1 ounce.

Type II.—Soft rubber tip.
- Style A.—Cone tip.
  - Capacities:
    - ¼ ounce.
    - ½ ounce.
    - 1 ounce.
- Style B.—Tapered tip.
  - Capacities:
    - 1 ounce.
    - 2 ounces.
    - 3 ounces.
    - 4 ounces.

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C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—

C–1a. *Glass.*—The glass shall be good quality clear glass tubing, as specified hereinafter.

C–1b. *Rubber.*—The rubber shall be natural or synthetic rubber suitable for the intended purpose.

C–2. *Workmanship.*—The workmanship shall be such that syringes will be free from defects which affect their appearance or which may impair their serviceability.

D. GENERAL REQUIREMENTS

D–1. *Bulb.*—The bulb shall be spherical in shape, molded with a smooth inner surface, and shall have a cylindrical neck of heavy reinforced rubber. The neck of the bulb shall fit tightly within the flared end of the barrel to insure against leakage during suction and compression. The bulb shall be capable of withstanding successive sterilizations in boiling water or by autoclaving, without damage. After compression, the bulb shall be capable of filling not less than two-thirds of the glass barrel with glycerine, U. S. P., at 80°F. by suction, and shall completely empty the barrel with one steady compression.

D–2. *Resistance to aging.*—Rubber parts shall show no softening and shall not become brittle, sticky, or tacky when subjected to the accelerated aging test described in paragraph F 3a.

D–3. *Annealing and boiling.*—The glass tubing shall be of such composition and so annealed that it will show no ring strain when subjected to the annealing test described in paragraph F–3b; and shall show no corrosion, scumming, chipping, or cracking when subjected to the boiling test described in paragraph F–3c.

D–4. *Thermal shock resistance.*—The barrels shall not crack or break when subjected to the thermal shock test described in paragraph F–3d.

D–5. *Resistance to air pressure.*—The syringes shall show no leakage when subjected to the air pressure test described in paragraph F–3e.

D–6. *Marking.*—The capacity of the syringe and the manufacturer's name or registered trade-mark shall be molded into the bulb.

D–7. *Condition on delivery.*—All syringes when delivered shall be clean and ready for use.

E. DETAIL REQUIREMENTS

E–1. *Type I, glass tip.*—The glass-tip syringes shall consist essentially of a glass barrel fitted with a spherical rubber bulb. The syringes shall be of the design illustrated in figure 1 for each style and shall conform to the approximate dimensions shown in table I.

E–1a. *Style A, cone tip.*—

E–1a(1). *Barrel.*—The barrel shall be a clear glass, cylindrically shaped tube. The distal end of the barrel shall be uniformly drawn to a smoothly finished tip; the proximal end shall be flared to provide a proper seating for the bulb, with the edge of the barrel everted to form a lip.

E–1a(2). *Cone tip.*—The cone-shaped tip shall be flame glazed.

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E–1b. *Style B, tapered tip.*—

E–1b(1). *Barrel.*—The barrel shall conform to the requirements in paragraph E–1a(1).

E–1b(2). *Tapered tip.*—The taper of the tip shall be gradual and consistent. Constriction and drawings shall cause no appreciable thinning of the wall. Edges shall be well rounded.

E–2. *Type II, soft rubber tip.*—The rubber-tip syringes shall consist essentially of a glass barrel with spherical rubber bulb, and a soft rubber tip so constructed as to avoid injury to the patient. The syringes shall be of the design illustrated in figure 1 for each style and shall conform to the approximate dimensions shown in table I.

E–2a. *Style A, cone tip.*—

E–2a(1). *Barrel.*—The barrel shall be a clear glass cylindrically-shaped tube. The distal end shall be drawn to form a shoulder for receiving the rubber cone tip, and the proximal end shall be flared to receive the bulb. When necessary for proper seating of the bulb, the proximal end shall have a constricted area.

E–2a(2). *Cone tip.*—The cone tip shall be of rubber, and shall fit snugly and securely on the shoulder of the barrel.

E–2b. *Style B, tapered tip.*—

E–2b(1). *Barrel.*—The barrel shall have a constricted area at the hub approximately 0.025 inch in length, with walls practically parallel.

E–2b(2). *Tapered tip.*—The tapered tip shall be as specified in paragraph E–1b(2); the rubber tip shall be removable.

E–2b(3). *Marking (for 1-ounce capacity only.)*—Type II, style B syringes of 1-ounce capacity shall be permanently marked "Syringe, Dakin's Solution".

TABLE I.—*Dimensions of glass syringes*

Type	Capacity	Barrel			
		Length		Inside diameter at proximal end	Wall thickness, minimum
		Style A, cone tip	Style B, tapered tip		
	Ounces	*Inches*	*Inches*	*Inches*	*Inches*
I, glass tip	⅛	3.125	.....	0.375	0.03
	¼	4	.....	0.50	0.03
	½	5.125	5.250	0.625	0.04
	1	5.375	5.875	0.75	0.04
II, soft rubber tip	½	4.125	.....	0.47	0.03
	½	4.75	.....	0.625	0.05
	1	4.625	4.75	0.75	0.05
	2	.....	7.50	0.86	0.05
	3	.....	7.50	0.94	0.05
	4	.....	8	0.94	0.05

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—Unless otherwise specified, at least 10 syringes of each type, style, and capacity shall be taken at random from each delivery for inspection and tests.

F–2. *Inspection.*—Samples shall be inspected for conformance to the requirements of this specification with respect to material, workmanship, dimensions, and marking.

F–3. *Tests.*—

F–3a. *Accelerated aging test.*—Sample bulbs and rubber tips shall be subjected to the accelerated aging test described in Federal Specification ZZ–R–601, using the oxygen bomb method at 70° ± 1°C., 300 ± 10 pounds per square inch, for a period of 7 days.

F–3b. *Annealing test.*—The glass barrels shall be viewed in a polariscope to determine conformance to the requirements in paragraph D–3.

F–3c. *Boiling test.*—The barrels taken for test shall be completely immersed in distilled water in a closed vessel having a small steam vent, and boiled continuously for 6 hours, boiling distilled water being added occasionally to compensate for evaporation losses. The barrels shall be suspended in water so that they will not come in contact with the walls of the containing vessel.

F–3d. *Thermal shock test.*—The barrels taken for test shall be immersed in tap water at 18.5° ±2.5°C. (65° ± 5°F.), allowed to remain 10 minutes, then transferred suddenly into boiling water. This operation shall be repeated five times.

F–3e. *Air pressure test.*—The syringes, completely assembled, shall be subjected to 20 pounds of air pressure.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

G–3. *Marking.*—

G–3a. *Packages.*—Each cardboard box or carton shall be plainly labeled with the name of the item, type, style, capacity, quantity contained, and the name of the manufacturer or contractor.

G–3b. *Shipping containers.*—Unless otherwise specified, the shipping containers shall be marked with the name of the item, the type, style, capacity, and quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments:

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H-2. *Army and Navy.*—

H-2a. *Applicable specifications and other publications.*—

Joint Army-Navy Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100–2—Standard Specification for Marking Shipments by Contractors.[1]

Navy Department Specifications:

General Specifications for Inspection of Material.[2]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[2]

H-2a(1). Copies of Federal and Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department. Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Federal, Joint Army-Navy, and U. S. Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-2a(2). Copies of Federal and Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Norfolk, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-2b. *Packaging, interior.*—

H-2b(1). *Unit packaging.*—Each unit shall be packaged in a set-up, paperboard box of appropriate size, conforming to the requirements of Joint Army-Navy Specification JAN–P–133. Boxes shall contain sufficient cushioning material to prevent shifting or breakage.

H-2b(2). *Intermediate packaging.*—

H-2b(2)a. *Domestic.*—Twelve packaged units shall be further packaged in a folding paperboard carton, or set-up paperboard

[1] Applicable to Army purchases only.
[2] Applicable to Navy purchases only.

box, conforming to the requirements of Joint Army-Navy Specifications JAN–P–120 and JAN–P–133, respectively.

H-2b(2)b. *Overseas.*—Twelve units shall be packaged as described in paragraph H-2b(2)a, and in addition shall be overwrapped with waterproof paper conforming to type C–1 of Joint Army-Navy Specification JAN–P–125, and sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140.

H-2c. *Packing, exterior.*—

H-2c(1). *Domestic.*—Unless otherwise specified, the following quantities shall be packed in a nailed wood box of appropriate size, conforming to the requirements of Joint Army-Navy Specification JAN–P–106, style 2, 2½, or 4, type 2 load (except that the requirements for container manufacturer's identification shall not apply); closure and strapping shall conform to the appendix thereto. The gross weight shall not exceed approximately 200 pounds.

Capacities	Number of intermediate packages to shipping container	Number of unit packages to shipping container
Ounces		
⅛	48	576
¼	30	360
½	24	288
1	12	144
2, 3, and 4	6	72

H-2c(2). *Overseas.*—Unless otherwise specified, the commodity shall be packed as described in paragraph H-2c(1), and in addition each box shall be lined with a sealed waterproof barrier, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN–P–125 for case liners; seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140.

H-2d. *Marking.*—

H-2d(1). *Unit packages and intermediate packages.*—Each package, barrier, and carton shall be marked with the stock numbers, nomenclature, unit, quantity, and name of manufacturer.

H-2d(2). *Shipping containers.*—

H-2d(2)a. *Army.*—Shipping containers shall be marked in accordance with the requirements of U. S. Army Specification 100–2, and shall bear the Joint Army-Navy stock number and the Army stock number.

H-2d(2)b. *Navy.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook, and shall bear the Joint Army-Navy stock number.

H-2e. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packaged, packed, and shipping containers marked for domestic or overseas shipment. (See pars. H-2b, H-2c, and H-2d.)

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I. NOTES

I–1. Purchasers should specify type, style, and capacity required, and should exercise any desired options offered herein. (See pars. B–1, F–1, G–1, G–2, G–3b, H–2c, and H–2e.)

I–2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–3. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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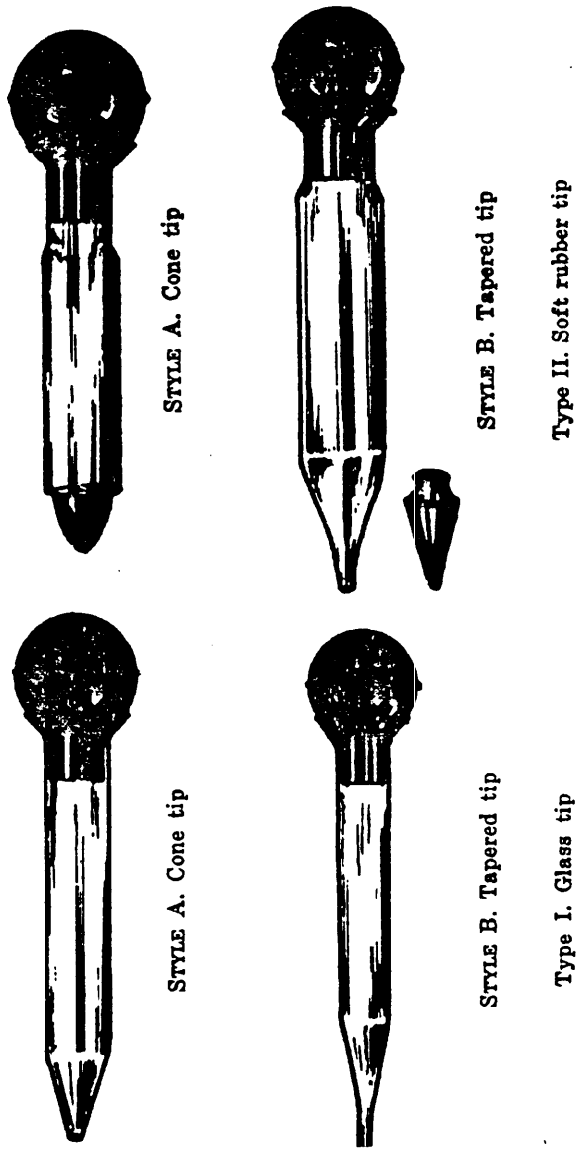


FIGURE 1.—Glass syringes, with bulb

Copies of this specification may be purchased Price 5 cents.

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